

General Overview of STC15W401AS series MCU

1. Introduction of STC15W401AS series MCU (In abundant supply)

STC15W401AS series MCU is a single-chip microcontroller based on a high performance 1T architecture 8051 CPU, which is produced by STC MCU Limited. It is a new generation of 8051 MCU of high speed, high stability, wide voltage range, low power consumption and super strong anti-disturbance. Besides, STC15W401AS series MCU is a MCU of super advanced encryption, because it adopts the ninth generation of STC encryption technology. With the enhanced kernel, STC15W401AS series MCU is faster than a traditional 8051 in executing instructions (about 8~12 times the rate of a traditional 8051 MCU), and has a fully compatible instruction set with traditional 8051 series microcontroller. External expensive crystal can be removed by being integrated internal high-precise R/C clock($\pm 0.3\%$) with $\pm 1\%$ temperature drift ($-40^{\circ}\text{C}\sim+85^{\circ}\text{C}$) while $\pm 0.6\%$ in normal temperature ($-20^{\circ}\text{C}\sim+65^{\circ}\text{C}$) and wide frequency adjustable between 5MHz and 35MHz. External reset circuit also can be removed by being integrated internal highly reliable one with 16 levels optional threshold voltage of reset. The STC15W401AS series MCU retains all features of the traditional 8051. In addition, it has 3-channels CCP/PCA/PWM, 8-channels and 10-bits A/D Converter(300 thousand times per sec.), a high-speed asynchronous serial port---UART(can be regarded as 3 serial ports by shifting among 3 groups of pins) and a high-speed synchronous serial peripheral interface---SPI. STC15W401AS series MCU is usually used in communications which need for several UARTs or electrical control or some occasion with strong disturbance.

In Keil C development environment, select the Intel 8052 to compiling and only contain < reg51.h > as header file.

STC15 series MCU with super high-speed CPU core of STC-Y5 works 20% faster than STC early 1T series (such as STC12/STC11/STC10 series) at same clock frequency.

- Enhanced 8051 Central Processing Unit, 1T, single clock per machine cycle, faster 8~12 times than the rate of a traditional 8051.
- Operating voltage range: 5.5V ~ 2.4V.
- On-chip 4K/8K/10K/12K/13K/15.5K FLASH program memory with flexible ISP/IAP capability, can be repeatedly erased more than 100 thousand times.
- on-chip 512 bytes SRAM: 256 byte scratch-pad RAM and 256 bytes of auxiliary RAM
- On-chip EEPROM with large capacity can be repeatedly erased more than 100 thousand times.
- ISP/IAP, In-System-Programming and In-Application-Programming , no need for programmer and emulator.
- 8 channels and 10 bits Analog-to-Digital Converter (ADC), the speed up to 300 thousand times per second, 3 channels PWM also can be used as 3 channels D/A Converter(DAC).
- 3 channels Capture/Compare units(CCP/PCA/PWM)
---- can be used as 3 Times or 3 external Interrupts(can be generated on rising or falling edge) or 3 channels D/A Converter.
- The high-speed pulse function of CCP/PCA can be utilized to realize 3 channels 9 ~ 16 bit PWM (each channel of which takes less than 0.6% system time)
- The clock output function of T0, T1 or T2 can be utilized to realize 8 ~ 16 bit PWM with a high degree of accuracy (which takes less than 0.4% system time)
- Internal highly reliable Reset with 16 levels optional threshold voltage of reset, external reset circuit can be completely removed

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- Internal high- precise R/C clock($\pm 0.3\%$) with $\pm 1\%$ temperature drift ($-40^{\circ}\text{C}\sim+85^{\circ}\text{C}$) while $\pm 0.6\%$ ($-20^{\circ}\text{C}\sim+65^{\circ}\text{C}$) in normal temperature and wide frequency adjustable between 5MHz and 35MHz (5.5296MHz / 11.0592MHz / 22.1184MHz / 33.1776MHz)
- No need external crystal and reset, and can output clock and low reset signal from MCU.
- Operating frequency range: 0- 35MHz, is equivalent to traditional 8051:0~420MHz.
- A high-speed asynchronous serial port----UART (can be regarded as 3 serial ports by shifting among 3 groups of pins):
UART(RxD/P3.0, TxD/P3.1) can be switched to (RxD_2/P3.6, TxD_2/P3.7),
also can be switched to (RxD_3/P1.6, TxD_3/P1.7).
- A high-speed synchronous serial peripheral interface----SPI.
- Support the function of Encryption Download (to protect your code from being intercepted).
- Support the function of RS485 Control
- Code protection for flash memory access, excellent noise immunity, very low power consumption
- Power management mode: Slow-Down mode, Idle mode(all interrupt can wake up Idle mode), Stop/Power-Down mode.
- Timers which can wake up stop/power-down mode: have internal low-power special wake-up Timer.
- Resource which can wake up stop/power-down mode are: INT0/P3.2, INT1/P3.3 (INT0/INT1, may be generated on both rising and falling edges), INT2/P3.6, INT3/P3.7, INT4/P3.0 (INT2/INT3/INT4, only be generated on falling edge); pins CCP0/CCP1/CCP2; pins RxD; pins T0/T2(their falling edge can wake up if T0/T2 have been enabled before power-down mode, but no interrupts can be generated); internal low-power special wake-up Timer.
- Five Timers/Counters, two 16-bit reloadable Timer/Counter(T0/T2, T0 is compatible with Timer0 of traditional 8051), T0/T2 all can independently achieve external programmable clock output, 3 channels CCP/PWM/PCA also can be used as three timers.
- Programmable clock output function(output by dividing the frequency of the internal system clock or the input clock of external pin):
The speed of external programmable clock output of 5V MCU is also not more than 13.5MHz, because the output speed of I/O port of STC15 series 5V MCU is not more than 13.5MHz.
The speed of external programmable clock output of 3.3V MCU is also not more than 8MHz, because the output speed of I/O port of STC15 series 3.3V MCU is not more than 8MHz.
① The Programmable clock output of T0 is on P3.5/T0CLKO (output by dividing the frequency of the internal system clock or the input clock of external pin T0/P3.4)
② The Programmable clock output of T2 is on P3.0/T2CLKO (output by dividing the frequency of the internal system clock or the input clock of external pin T2/P3.1)
Two timers/counters in above all can be output by dividing the frequency from 1 to 65536.
③ The Programmable clock output of master clock is on P5.4/SysClkO, and its frequency can be divided into SysClk/1, SysClk/2, SysClk/4.

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The master clock can either be internal R/C clock or the external input clock or the external crystal oscillator.

SysClk is the frequency of master clock. SysClkO is the output of master clock.

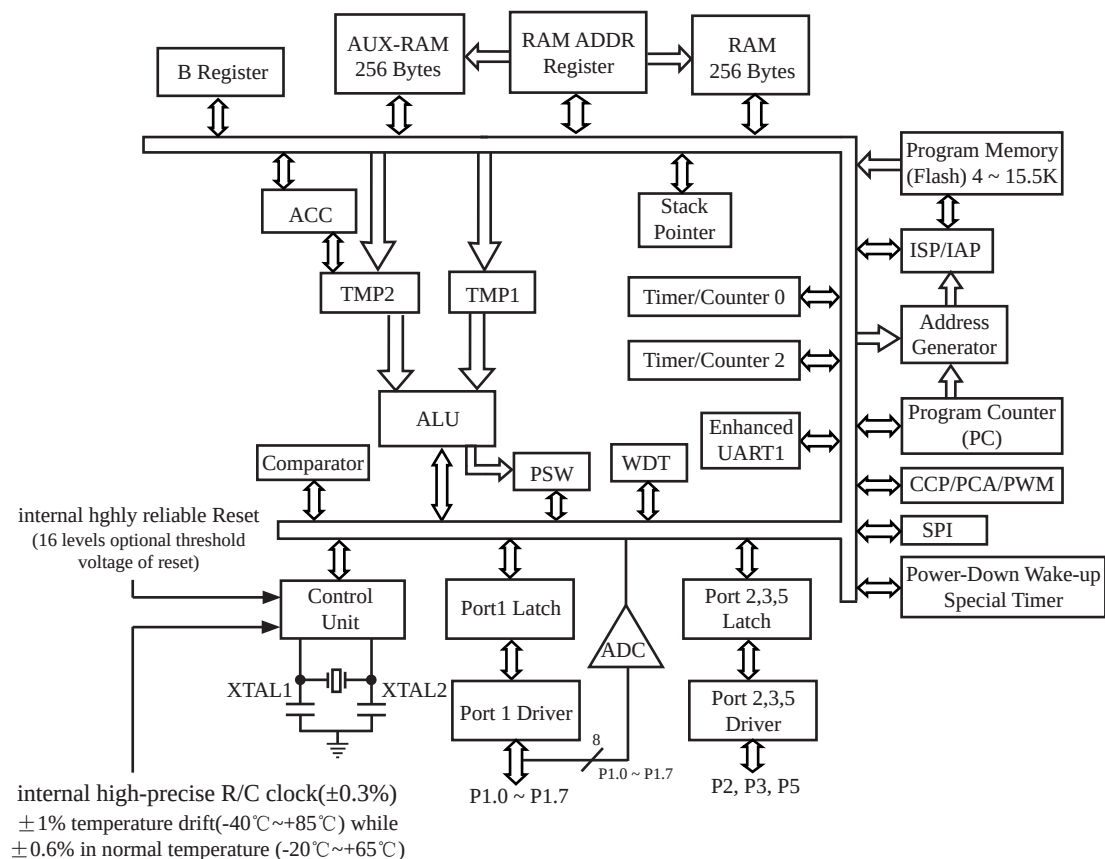
- **Comparator**, which support comparing by external pin CMP+ and CMP- or internal reference voltage and generating output signal (its polarity can be configured) on CMPO pin can be used as 1 channel ADC or brownout detect function.
- One 15 bits Watch-Dog-Timer with 8-bit pre-scaler (one-time-enabled)
- advanced instruction set, which is fully compatible with traditional 8051 MCU, have hardware multiplication / division command.
- 26/18/14 common I/O ports are available, their mode is quasi_bidirectional/weak pull-up (traditional 8051 I/O ports mode) after reset, and can be set to four modes: quasi_bidirectional/weak pull-up, strong push-pull/strong pull-up, input-only/high-impedance and open drain.
the driving ability of each I/O port can be up to 20mA, but the current of the whole chip don't exceed this maximum 90mA.

If I/O ports are not enough, it can be extended by connecting a 74HC595(reference price: RMB 0.15 yuan). Besides, cascading several chips also can extend to dozens of I/O ports.

- Package: SOP28, TSSOP28 (6.4mm x 9.7mm), QFN28 (5mm x 5mm), SKDIP28, SOP20, DIP20, TSSOP20(6.5mm x 6.5mm), SOP16, DIP16.
- All products are baked 8 hours in high-temperature 175°C after be packaged, Manufacture guarantee good quality.
- In Keil C development environment, select the Intel 8052 to compiling and only contain < reg51.h > as header file.

2. Block diagram of STC15W401AS series

The internal structure of STC15W401AS series MCU is shown in the block diagram below. STC15W401AS series MCU includes central processor unit(CPU), program memory (Flash), data memory(SRAM), Timers/ Counters, power-down wake-up Timer, I/O ports, high-speed A/D converter(ADC), Comparator, Watchdog, high-speed asynchronous serial communication ports---UART, CCP/PWM/PCA, a group of high-speed synchronous serial peripheral interface (SPI), internal high- precise R/C clock, internal highly reliable Reset and so on. STC15W401AS series MCU almost includes all of the modules required in data acquisition and control, and can be regarded as an on-chip system (SysTem Chip or SysTem on Chip, abbreviated as STC, this is the name origin of Hongjing technology STC Limited).

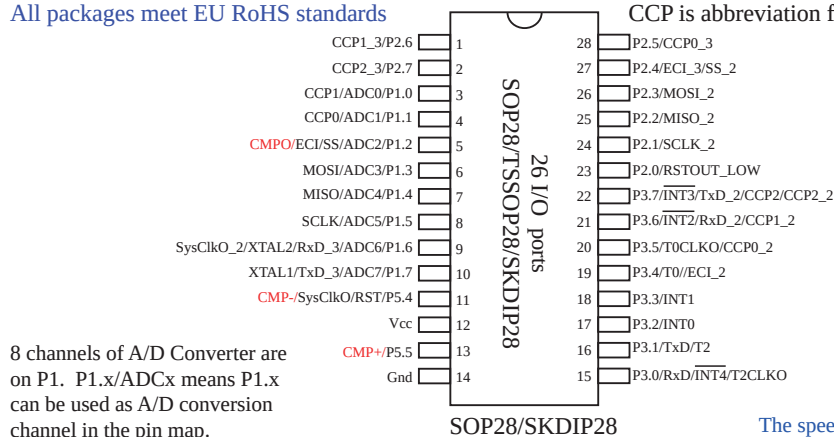


STC15W401AS series Block Diagram

3. Pin Configurations of STC15W401AS series MCU

All packages meet EU RoHS standards

CCP is abbreviation for Capture, Compare, PWM



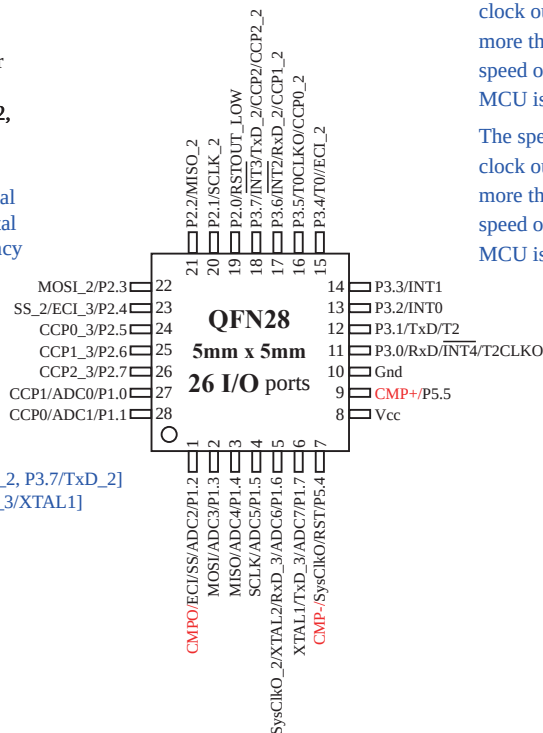
SysClkO is the output of master clock whose frequency can be divided into SysClk/1, SysClk/2, SysClk/4

The master clock can either be internal R/C clock or the external input clock or the external crystal oscillator. SysClk is the frequency of master clock.

The speed of external programmable clock output of 5V MCU is also not more than 13.5MHz, because the output speed of I/O port of STC15 series 5V MCU is not more than 13.5MHz.

The speed of external programmable clock output of 3.3V MCU is also not more than 8MHz, because the output speed of I/O port of STC15 series 3.3V MCU is not more than 8MHz.

Recommend UART1 on [P3.6/RxD_2, P3.7/TxD_2] or [P1.6/RxD_3/XTAL2, P1.7/TxD_3/XTAL1]

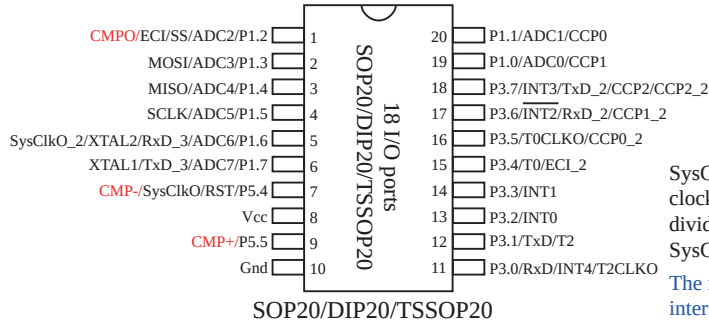


T0CLKO refers to the programmable clock output of Timer/Counter 0 (output by dividing the frequency of the internal system clock or the input clock of external pin T0/P3.4);

T2CLKO refers to the programmable clock output of Timer/Counter 2 (output by dividing the frequency of the internal system clock or the input clock of external pin T2/P3.1);

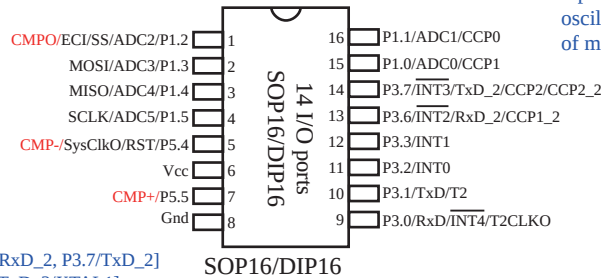
In addition to programmable output on the internal system clock, T0CLKO/T2CLKO also can be used as divider by dividing the frequency of the internal system clock or the input clock of external pin T0/T2.

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SysClkO is the output of master clock whose frequency can be divided into SysClk/1, SysClk/2, SysClk/4

The master clock can either be internal R/C clock or the external input clock or the external crystal oscillator. SysClk is the frequency of master clock.



Recommend UART1 on [P3.6/RxD_2, P3.7/TxD_2] or [P1.6/RxD_3/XTAL2, P1.7/TxD_3/XTAL1]

Mnemonic	Add	Name	7	6	5	4	3	2	1	0	Reset Value
AUXR1 P_SW1	A2H	Auxiliary register 1	S1_S1	S1_S0	CCP_S1	CCP_S0	SPI_S1	SPI_S0	0	DPS	00x0,x00x
CLK_DIV (PCON2)	97H	Clock Division register	MCKO_S1	MCKO_S0	ADRJ	Tx_Rx	MCLKO_2	CLKS2	CLKS1	CLKS0	0000,0000

UART1/S1 can be switched in 3 groups of pins by selecting the control bits S1_S0 and S1_S1.

S1_S1	S1_S0	UART1/S1 can be switched between P1 and P3
0	0	UART1/S1 on [P3.0/RxD,P3.1/TxD]
0	1	UART1/S1 on [P3.6/RxD_2,P3.7/TxD_2]
1	0	UART1/S1 on [P1.6/RxD_3/XTAL2,P1.7/TxD_3/XTAL1] when UART1 is on P1, please using internal R/C clock.
1	1	Invalid

Recommend UART1 on [P3.6/RxD_2, P3.7/TxD_2] or [P1.6/RxD_3/XTAL2, P1.7/TxD_3/XTAL1].

CCP can be switched in 3 groups of pins by selecting the control bits CCP_S1 and CCP_S0.

CCP_S1	CCP_S0	CCP can be switched in P1 and P3
0	0	CCP on [P1.2/ECI,P1.1/CCP0,P1.0/CCP1,P3.7/CCP2]
0	1	CCP on [P3.4/ECI_2,P3.5/CCP0_2,P3.6/CCP1_2,P3.7/CCP2_2]
1	0	CCP on [P2.4/ECI_3,P2.5/CCP0_3,P2.6/CCP1_3,P2.7/CCP2_3]
1	1	Invalid

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Mnemonic	Add	Name	7	6	5	4	3	2	1	0	Reset Value
AUXR1 P_SW1	A2H	Auxiliary register 1	S1_S1	S1_S0	CCP_S1	CCP_S0	SPI_S1	SPI_S0	0	DPS	00x0,x00x
CLK_DIV (PCON2)	97H	Clock Division register	SysCKO_S1	SysCKO_S0	ADJR	Tx_Rx	SysClkO_2	CLKS2	CLKS1	CLKS0	0000,0000

SPI can be switched in 2 groups of pins by selecting the control bit SPI_S0

SPI_S1	SPI_S0	SPI can be switched in P1 and P2
0	0	SPI on [P1.2/SS,P1.3/MOSI,P1.4/MISO,P1.5/SCLK]
0	1	SPI on [P2.4/SS_2,P2.3/MOSI_2,P2.2/MISO_2,P2.1/SCLK_2]
1	0	SPI on [P5.4/SS_3,P4.0/MOSI_3,P4.1/MISO_3,P4.3/SCLK_3]
1	1	Invalid

SysCKO_S1	SysCKO_S0	the control bit of master clock output by dividing the frequency (The master clock can either be internal R/C clock or the external input clock or the external crystal oscillator)
0	0	Master clock do not output external clock
0	1	Master clock output external clock, but its frequency do not be divided, and the output clock frequency = SysClk / 1
1	0	Master clock output external clock, but its frequency is divided by 2, and the output clock frequency = SysClk / 2
1	1	Master clock output external clock, but its frequency is divided by 4, and the output clock frequency = SysClk / 4

The master clock can either be internal R/C clock or the external input clock or the external crystal oscillator.
SysClk is the frequency of master clock.

STC15W401AS series MCU output master clock on SysClkO/P5.4

SysClkO_2: to select Master Clock output on where

0: Master Clock output on SysClkO/P5.4

1: Master Clock output on SysClkO_2/XTAL2/P1.6

The master clock can either be internal R/C clock or the external input clock or the external crystal oscillator.

ADJR: the adjustment bit of ADC result

0: ADC_RES[7:0] store high 8-bit ADC result, ADC_RES[1:0] store low 2-bit ADC result

1: ADC_RES[1:0] store high 2-bit ADC result, ADC_RES[7:0] store low 8-bit ADC result

Tx_Rx: the set bit of relay and broadcast mode of UART1

0: UART1 works on normal mode

1: UART1 works on relay and broadcast mode, that to say output the input level state of RxD port to the outside TxD pin in real time, namely the external output of TxD pin can reflect the input level state of RxD port.

the RxD and TxD of UART1 can be switched in 3 groups of pins: [RxD/P3.0, TxD/P3.1];

[RxD_2/P3.6, TxD_2/P3.7];

[RxD_3/P1.6, TxD_3/P1.7].

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Mnemonic	Add	Name	7	6	5	4	3	2	1	0	Reset Value
CLK_DIV (PCON2)	97H	Clock Division register	MCKO_S1	MCKO_S0	ADRJ	Tx_Rx	MCLKO_2	CLKS2	CLKS1	CLKS0	0000,0000

CLKS2	CLKS1	CLKS0	the control bit of system clock (System clock refers to the master clock that has been divided frequency, which is offered to CPU, UARTs, SPI, Timers, CCP/PWM/PCA and A/D Converter)
0	0	0	Master clock frequency/1, No division
0	0	1	Master clock frequency/2
0	1	0	Master clock frequency/4
0	1	1	Master clock frequency/8
1	0	0	Master clock frequency/16
1	0	1	Master clock frequency/32
1	1	0	Master clock frequency/64
1	1	1	Master clock frequency/128

The master clock can either be [internal R/C clock](#) or the [external input clock](#) or the [external crystal oscillator](#).

4. STC15W401AS series Selection and Price Table

Type 1T 8051 MCU	Operating Voltage (V)	Flash (byte)	SRAM (byte)	UART	5 Timers T0/T2	CCP PCA PWM	Special Power- down Wake-up Timer	Standard External Interrupts	A/D 8-channel	COM PAR ATR OR	EEP ROM	Internal Low- Voltage Detection Interrupt	W D T	Internal High- reliable Reset (with optional threshold voltage)	Internal High- Precise Clock	Output clock and reset signal from MCU	Encryption Download (to protect your code from being intercepted)	RS485 Control	All Packages SOP28/TSSOP28/ SKDIP28/QFN28 SOP20 / DIP20 / TSSOP20 SOP16 / DIP16	Price of a part of packages(RMB ¥)	SOP16	SOP20	SOP28
STC15W401AS series MCU Selection and Price Table Note: 3 channels CCP/PCA/PWM also can be used as 3 Timers.																							
STC15W401AS	2.4-5.5	1K	512	1	Y	2	3-ch	Y	5	10-bit	Y	1	5K	Y	Y	16-level	Y	Y	Y	Y			
STC15W402AS	2.4-5.5	2K	512	1	Y	2	3-ch	Y	5	10-bit	Y	1	5K	Y	Y	16-level	Y	Y	Y	Y			
STC15W404AS	2.5-5.5	4K	512	1	Y	2	3-ch	Y	5	10-bit	Y	1	9K	Y	Y	16-level	Y	Y	Y	Y			
STC15W408AS	2.5-5.5	8K	512	1	Y	2	3-ch	Y	5	10-bit	Y	1	5K	Y	Y	16-level	Y	Y	Y	Y			
IAP15W413AS	2.5-5.5	13K	512	1	Y	2	3-ch	Y	5	10-bit	Y	1	IAP	Y	Y	16-level	Y	Y	Y	Y	The program Flash in user program area can be used as EEPROM.		
IRC15W415AS (Using external crystal or internal 24MHz clock)	2.5-5.5	15.5K	512	1	Y	2	3-ch	Y	5	10-bit	Y	1	IAP	Y	Y	Fixed	Y	Y	N	N	The program Flash in user program area can be used as EEPROM.		

Encryption Download : please burn source code with encryption key onto MCU in the factory. Then, you can make a simple update software just with one "update" button by firstly using the fuction "encrytion download" and then "release project" to update yourself code unabled to be intercepted when you need to upgrade your code.

To provide customized IC services

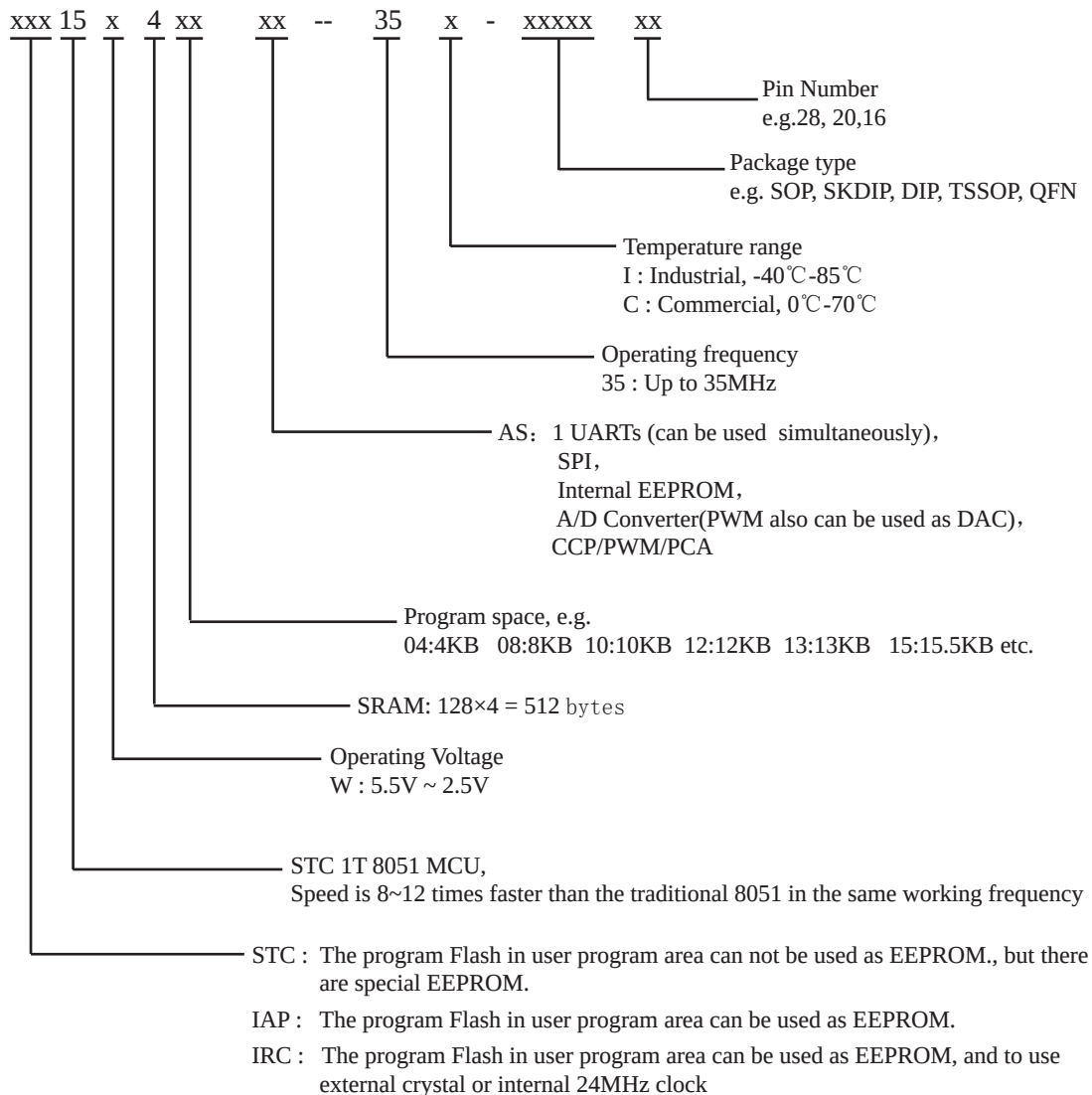
Because the last 7 bytes of the program area is stored mandatorily the contents of only global ID, the program space the user can actually use is 7 bytes smaller than the space shown in the selection table.

Conclusion : STC15W401AS series MCU have: Two16-bit relaodable Timers/Counters that are Timer/Counter 0 and Timer/Counter 2; 3 channels CCP/PWM/PCA (can achieve 3 timers or 3 D/A converters again); special power-down wake-up timer; 5 external interrupts INT0/INT1/INT2/INT3/INT4; a high-speed asynchronous serial port ---- UART; a high-speed synchronous serial peripheral interface ---- SPI; 8 channels and 10 bits high-speed A/D converter; 1 Comparator; 1 data pointers ---- DPTR.

5. STC15W401AS series Package and Price Table

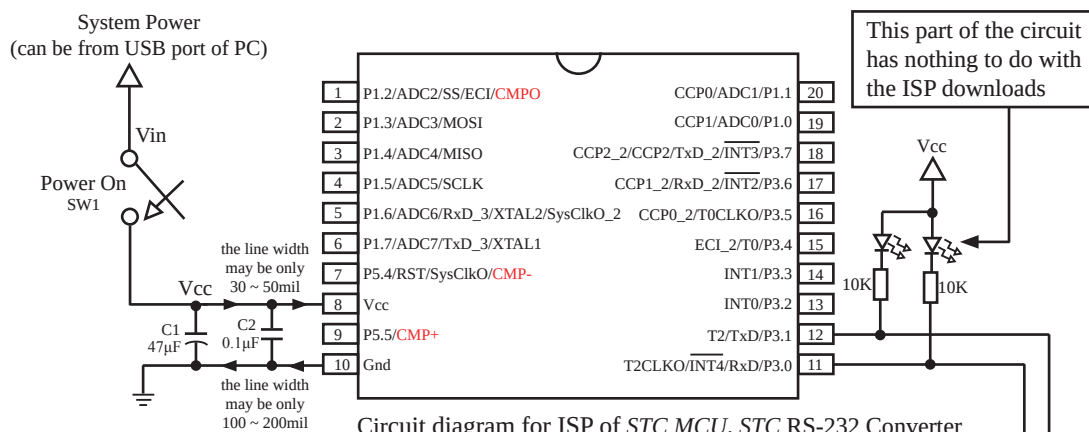
Type 1T 8051 MCU	Operating Voltage (V)	Operating Frequency (MHz)	Operating Temprature (I — Industrial)	All Packages Price(RMB ¥)									
				SOP28/ TSSOP28/ SKDIP28/ QFN28/ SOP20/ DIP20/ TSSOP20/ SOP16/ DIP16									
				SOP28	TSSOP28	SKDIP28	QFN28	SOP20	TSSOP20	DIP20	SOP16	DIP16	
STC15W401AS series MCU Package and Price Table													
STC15W401AS	2.5-5.5	35	-40℃ ~ +85℃										
STC15W402AS	2.5-5.5	35	-40℃ ~ +85℃										
STC15W404AS	2.5-5.5	35	-40℃ ~ +85℃										
STC15W401AS	2.5-5.5	35	-40℃ ~ +85℃										
IAP15W413AS	2.5-5.5	35	-40℃ ~ +85℃										
IRC15W415AS	2.5-5.5	35	-40℃ ~ +85℃										

6. Naming rules of STC15W401AS series MCU



7. Application Circuit Diagram for ISP of STC15W401AS series MCU

7.1 Application Circuit Diagram for ISP using RS-232 Converter



Please power on the target MCU after press down the button "Download/Program" on STC-ISP.exe when burning code to MCU.

Internal highly reliable Reset, External reset circuit can be completely removed.

P5.4/RST/MCLKO pin factory defaults to the I/O port, which can be set as RST reset pin(active high) through the STC-ISP programmer.

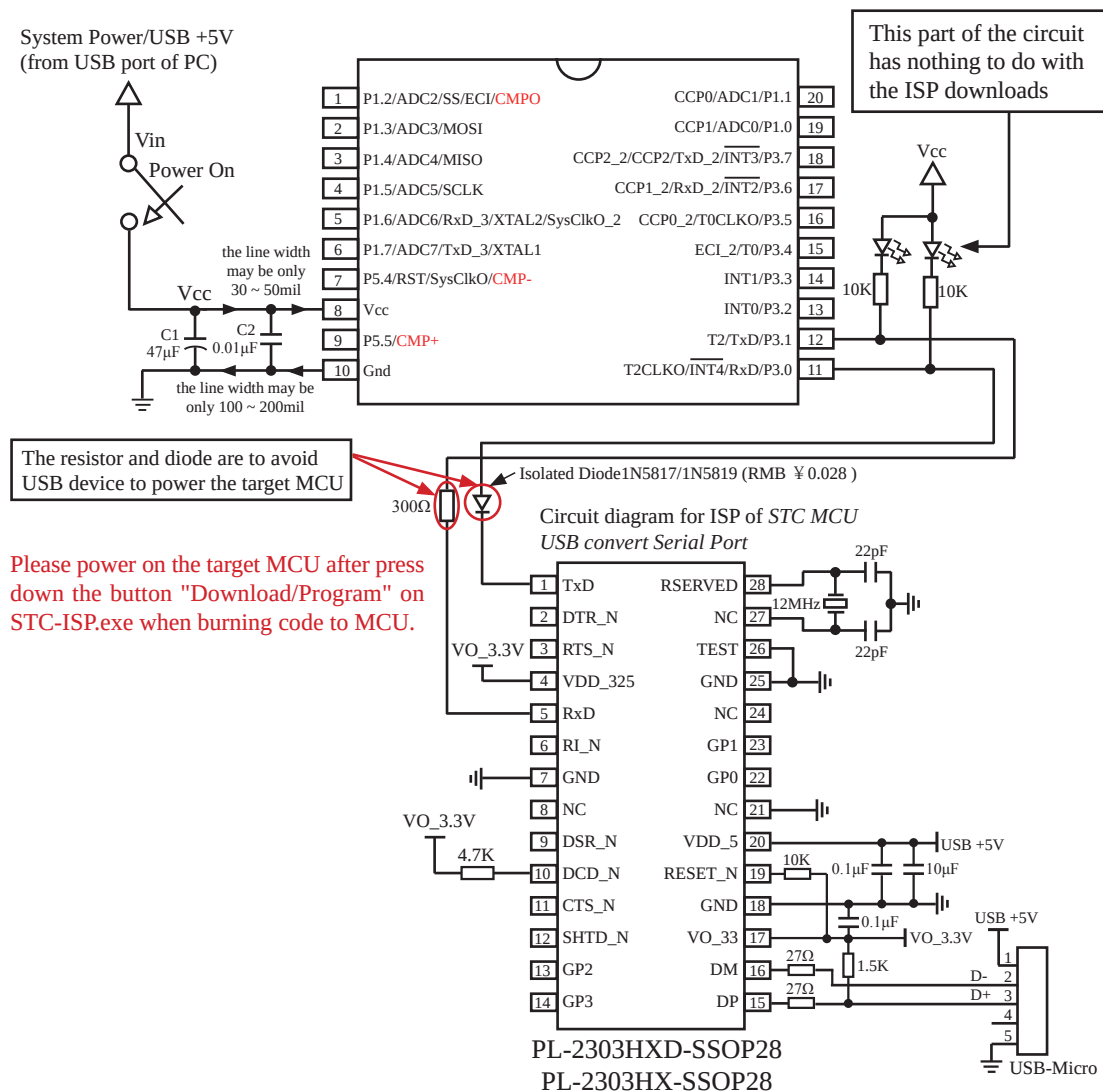
Internal high-precise R/C clock($\pm 3\%$), $\pm 1\%$ temperature drift ($-40^{\circ}\text{C} \sim +85^{\circ}\text{C}$) while $\pm 0.6\%$ in normal temperature ($-20^{\circ}\text{C} \sim +65^{\circ}\text{C}$). External expensive crystal can be completely removed.

Recommend to add decoupling capacitor C1(47µF) and C2(0.1µF) between Vcc and Gnd that can remove power noise and improve the anti-interference ability.



General Overview of STC15W401AS series MCU

7.3 Application Circuit Diagram for ISP using USB Chip PL-2303HXD / PL-2303HX to convert Serial Port



Internal highly reliable Reset, External reset circuit can be completely removed.

P5.4/RST/MCLKO pin factory defaults to the I/O port, which can be set as RST reset pin(active high) through the STC-ISP programmer.

Internal high-precise R/C clock(±3%), ±1% temperature drift (-40℃~+85℃) while ±0.6% in normal temperature (-20℃~+65℃). External expensive crystal can be completely removed.

Recommend to add decoupling capacitor C1(47μF) and C2(0.1μF) between Vcc and Gnd that can remove power noise and improve the anti-interference ability.

8. Pin Descriptions of STC15W401AS series MCU

MNEMONIC	Pin Number				DESCRIPTION	
	SOP28 TSSOP28 SKDIP28	QFN28	SOP20 DIP20 TSSOP20	SOP16 DIP16		
P1.0/ADC0/ CCP1	3	27	19	15	P1.0	common I/O port PORT1[0]
					ADC0	ADC input channel-0
					CCP1	Capture of external signal(measure frequency or be used as external interrupts)、high-speed Pulse and Pulse-Width Modulation output channel-1
P1.1/ADC1/ CCP0	4	28	20	16	P1.1	common I/O port PORT1[1]
					ADC1	ADC input channel-1
					CCP0	Capture of external signal(measure frequency or be used as external interrupts)、high-speed Pulse and Pulse-Width Modulation output channel-0
P1.2/ADC2/SS/ ECI/CMPO	5	1	1	1	P1.2	common I/O port PORT1[2]
					ADC2	ADC input channel-2
					SS	Slave selection signal of synchronous serial peripheral interface----SPI
					ECI	External pulse input pin of CCP/PCA counter
					CMPO	The output port of reslut compared by comparator
P1.3/ADC3/ MOSI	6	2	2	2	P1.3	common I/O port PORT1[3]
					ADC3	ADC input channel-3
					MOSI	Master Output Slave Input of SPI
P1.4/ADC4/ MISO	7	3	3	3	P1.4	common I/O port PORT1[4]
					ADC4	ADC input channel-4
					MISO	Master Input Slave Output of SPI
P1.5/ADC5/ SCLK	8	4	4	4	P1.5	common I/O port PORT1[5]
					ADC5	ADC input channel-5
					SCLK	Clock Signal of synchronous serial peripheral interface---SPI
P1.6/ADC6/ Rx_D_3/XTAL2/ SysClkO_2	9	5	5		P1.6	common I/O port PORT1[6]
					ADC6	ADC input channel-6
					RxD_3	Receive Data Port of UART
					XTAL2	Output from the inverting amplifier of internal clock circuit. This pin should be floated when an external oscillator is used.
					SysClkO_2	Master clock output; the output frequency can be SysClk/1, SysClk/2 and SysClk/4. The master clock can either be internal R/C clock or the external input clock or the external crystal oscillator.
P1.7/ADC7/ Tx_D_3/XTAL1	10	6	6		P1.7	common I/O port PORT1[7]
					ADC7	ADC input channel-7
					TxD_3	Transit Data Port of UART
					XTAL1	Input to the inverting oscillator amplifier of internal clock circuit. Receives the external oscillator signal when an external oscillator is used.

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MNEMONIC	Pin Number				DESCRIPTION	
	SOP28 TSSOP28 SKDIP28	QFN28	SOP20 DIP20 TSSOP20	SOP16 DIP16		
P2.0/ RSTOUT_LOW	23	19			P2.0	common I/O port PORT2[0]
					RSTOUT_LOW	the pin output low after power-on and during reset, which can be set to output high by software
P2.1/SCLK_2	24	20			P2.1	common I/O port PORT2[1]
					SCLK_2	Clock Signal of synchronous serial peripheral interface----SPI
P2.2/MISO_2	25	21			P2.2	common I/O port PORT2[2]
					MISO_2	Master Input Slave Output of SPI
P2.3/MOSI_2	26	22			P2.3	common I/O port PORT2[3]
					MOSI_2	Master Output Slave Input of SPI
P2.4/ECI_3/SS_2	27	23			P2.4	common I/O port PORT2[4]
					ECI_3	External pulse input pin of CCP/PCA counter
					SS_2	Slave selection signal of synchronous serial peripheral interface----SPI
P2.5/CCP0_3	28	24			P2.5	common I/O port PORT2[5]
					CCP0_3	Capture of external signal(measure frequency or be used as external interrupts)、high-speed Pulse and Pulse-Width Modulation output channel-0
P2.6/CCP1_3	1	25			P2.6	common I/O port PORT2[6]
					CCP1_3	Capture of external signal(measure frequency or be used as external interrupts)、high-speed Pulse and Pulse-Width Modulation output channel-1
P2.7/CCP2_3	2	26			P2.7	common I/O port PORT2[7]
					CCP2_3	Capture of external signal(measure frequency or be used as external interrupts)、high-speed Pulse and Pulse-Width Modulation output channel-2
P3.0/RxD/ $\overline{\text{INT4}}$ / T2CLKO	15	11	11	9	P3.0	common I/O port PORT3[0]
					RxD	Receive Data Port of UART1
					$\overline{\text{INT4}}$	External interrupt 4, which only can be generated on falling edge. /INT4 supports power-down waking-up
P3.1/TxD/T2	16	12	12	10	T2CLKO	T2 Clock Output The pin can be configured for T2CLKO by setting INT_CLKO[2] bit /T2CLKO
					P3.1	common I/O port PORT3[1]
					TxD	Transit Data Port of UART1
P3.2/INT0	17	13	13	11	T2	External input of Timer/Counter 2
					P3.2	common I/O port PORT3[2]
					INT0	External interrupt 0, which both can be generated on rising and falling edge. INT0 only can generate interrupt on falling edge if IT0 (TCON.0) is set to 1. And, INT0 both can generate interrupt on rising and falling edge if IT0 (TCON.0) is set to 0.

General Overview of STC15W401AS series MCU

MNEMONIC	Pin Number				DESCRIPTION	
	SOP28 TSSOP28 SKDIP28	QFN28	SOP20 DIP20 TSSOP20	SOP16 DIP16		
P3.3/INT1	18	14	14	12	P3.3	common I/O port PORT3[3]
					INT1	External interrupt 1, which both can be generated on rising and falling edge. INT1 only can generate interrupt on falling edge if IT1 (TCON.2) is set to 1. And, INT1 both can generate interrupt on rising and falling edge if IT1 (TCON.2) is set to 0. INT1 supports power-down waking-up
P3.4/T0/ECI_2	19	15	15		P3.4	common I/O port PORT3[4]
					T0	External input of Timer/Counter 0
					ECI_2	External pulse input pin of CCP/PCA counter
P3.5/T0CLKO/ CCP0_2	20	16	16		P3.5	common I/O port PORT3[5]
					T0CLKO	T0 Clock Output The pin can be configured for T0CLKO by setting INT_CLKO[0] bit /T0CLKO
					CCP0_2	Capture of external signal(measure frequency or be used as external interrupts)、high-speed Pulse and Pulse-Width Modulation output channel-0
P3.6/ <u>INT2</u> /RxD_2 /CCP1_2	21	17	17	13	P3.6	common I/O port PORT3[6]
					<u>INT2</u>	External interrupt 2, which only can be generated on falling edge. /INT2 supports power-down waking-up
					RxD_2	Receive Data Port of UART1
					CCP1_2	Capture of external signal(measure frequency or be used as external interrupts)、high-speed Pulse and Pulse-Width Modulation output channel-1
P3.7/ <u>INT3</u> /TxD_2/ CCP2/CCP2_2	22	18	18	14	P3.7	common I/O port PORT3[7]
					<u>INT3</u>	External interrupt 3, which only can be generated on falling edge. INT3 supports power-down waking-up
					TxD_2	Transit Data Port of UART
					CCP2	Capture of external signal(measure frequency or be used as external interrupts)、high-speed Pulse and Pulse-Width Modulation output channel-2
					CCP2_2	Capture of external signal(measure frequency or be used as external interrupts)、high-speed Pulse and Pulse-Width Modulation output channel-2
P5.4/RST/ SysClkO/CMP-	11	7	7	5	P5.4	common I/O port PORT5[4]
					RST	Reset pin. A high on this pin for at least two machine cycles will reset the device.
					SysClkO	Master clock output; the output frequency can be SysClk/1, SysClk/2 and SysClk/4. The master clock can either be internal R/C clock or the external input clock or the external crystal oscillator.
					CMP-	Comparator negative input
P5.5/CMP+	13	9	9	7	P5.5	common I/O port PORT5[5]
					CMP+	Comparator positive input
Vcc	12	8	8	6	The positive pole of power	
Gnd	14	10	10	8	The negative pole of power, Ground	